

PARTH DINESHBHAI PANDYA

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SUMMARY

Software Engineer with strong experience in backend development using Python, Django, and FastAPI, along with a solid foundation in software design, object-oriented programming, and system-level concepts. Skilled in building scalable APIs, data processing systems, and clean, maintainable architectures. Familiar with multithreading, networking fundamentals (TCP/IP), and Linux/Windows environments. Experienced with SQL databases, cloud platforms like Azure, and CI/CD pipelines. Passionate about developing efficient, high-quality software solutions in collaborative and fast-paced environments.

EDUCATION

Master of Science in Computer & Information Systems Eng Tennessee State University CGPA: 3.33/4 Key Courses: System Modeling, Systems Eng, Software Systems Design, Operating Systems, AI, Digital Image Processing, Networks.	Aug 2024 - May 2026 Nashville, TN
Bachelors in information & technology Gujarat Technological University CGPA: 3.59/4	Aug 2017 – May 2021 Gujarat, India

TECHNICAL TOOLS

Languages & Frameworks: Python (Advanced, OOP, Design Patterns), Django, FastAPI, Flask
Backend & Architecture: REST API Development, Microservices, Distributed Systems, Event-Driven Architecture, Multithreading & Parallel Processing
Data & Computing: Pandas, NumPy, SciPy, Statistical Analysis, Time Series Analysis, Large-Scale Data Transformation
Databases: SQL (Advanced), PostgreSQL, Query Optimization, Indexing, Stored Procedures
Cloud & DevOps: AWS, Microsoft Azure, Docker, CI/CD Pipelines, Cloud Deployment & Monitoring
Integration: RESTful API Design, Third-Party API Integration, Data Validation & Error Handling
Practices : Agile (Scrum/Kanban), Code Reviews, Technical Documentation

EXPERIENCE

Teaching Assistant & Research Assistant Department of Electrical & Computer Engineering, Tennessee State University	08/2024 to 04/2026 USA
- Conduct technical and literature analysis across AI, sensor systems, and data-driven modeling, supporting system-level design decisions and iterative research development. - Design, train, and evaluate machine learning and statistical models using Python and MATLAB, focusing on correctness, performance, and reproducibility. - Implement data preprocessing, feature engineering, and visualization pipelines to ensure high-quality inputs and validated outputs. - Support system modeling, numerical analysis, and laboratory experimentation, including instrumentation setup and structured data acquisition workflows. - Write C-based programs using structured programming, memory management, and performance-aware techniques. - Assist with software-based data acquisition and backend integration, applying TCP/IP and UDP communication concepts relevant to distributed systems - Supported undergraduate courses in Probability & Statistics, Python Programming, and MATLAB-based simulation, reinforcing core computer science and problem-solving fundamentals. - Provided detailed grading and feedback to improve student understanding of algorithms, data analysis, and computational logic. - Delivered one-on-one and group tutoring on OOP concepts, numerical methods, and algorithmic thinking, adapting explanations to different learning styles. - Assisted during MATLAB lab sessions, debugging simulation issues and guiding students through modeling and validation workflows. - Collaborated with faculty to align instructional support with course goals and assessment criteria.	
Web Application Developer Stack Summation	07/2022 to 06/2024 India
- Designed and developed backend services using Python (Django) to support user-facing applications with reliable, scalable APIs. - Built and integrated RESTful APIs to process, validate, and serve data to applications, dashboards, and reporting workflows. - Developed additional backend services using FastAPI, focusing on high-performance API endpoints and clean, modular architecture. - Built Vue.js front-end components to display USA resort information, including seasonal weather insights and monthly snow probability for ski activities, presented through interactive charts and tabular views. - Implemented data visualization features (charts and tables) to improve user understanding of weather trends and activity availability. - Applied object-oriented design principles and wrote clean, maintainable code, actively participating in peer reviews and iterative improvements. - Optimized SQL queries (joins, aggregations, indexing) to improve application performance and data access latency by ~30%. - Integrated third-party APIs (Google Maps & Google Drive) to enable real-time features, geospatial data handling, and secure file storage for 500+ users. - Fixed critical backend (PHP) and frontend (Vue.js) bugs in a U.S.-based application, improving stability, usability, and performance. - Integrated PayPal payment gateway, handling secure transactions, callbacks, and error scenarios. - Diagnosed and resolved server-side issues, including fixing an automatic file upload process to ensure reliable deployment and data availability. - Supported production systems, including monitoring, debugging, and incident resolution using Git-based CI/CD pipelines. - Implemented logging, error handling, and data validation to improve observability, reliability, and fault tolerance. - Collaborated closely with product, QA, and engineering teams to deliver features on schedule while balancing quality, performance, and scalability trade-offs.	
Backend Developer Mindtrunk	07/2021 to 06/2022 India
- Developed and deployed real-time image analysis features on Raspberry Pi using Python, improving processing speed by 30% in edge device scenarios. - Designed and maintained database systems with SQLAlchemy, enabling seamless integration across SQLite, MySQL, and PostgreSQL engines. - Engineered custom computer vision solutions by extracting and processing image data with Qt5, enhancing accuracy and reliability in data interpretation. - Optimized backend performance for low-power environments, reducing latency and ensuring smooth API interactions on constrained hardware. - Actively participated in testing and debugging cycles, resolving hardware-software integration issues and contributing to stable and	

reliable deployments.

- Developed a responsive, cross-platform front-end using Vue.js, **improving UI/UX and increasing user engagement by 30%** across web and mobile applications.
- Integrated AWS S3 for video storage and retrieval, enabling seamless uploads and **reducing video load times by over 40%**.
- Built and optimized backend services using FastAPI, **enhancing data processing speed and reducing API response times by nearly 50%**.
- Designed and implemented RESTful APIs to facilitate smooth, secure communication between front-end and backend systems, supporting real-time event tracking.
- Conducted comprehensive scalability and load tests, ensuring system stability under heavy traffic and preparing the platform to **handle a 4x increase in concurrent users**.

CERTIFICATIONS

Certified in C – C-DAC, GIST (04/2018)

Certified in C++ – C-DAC, GIST (04/2019)